

Private Credit Due Diligence Agent

System Design

June 18, 2026

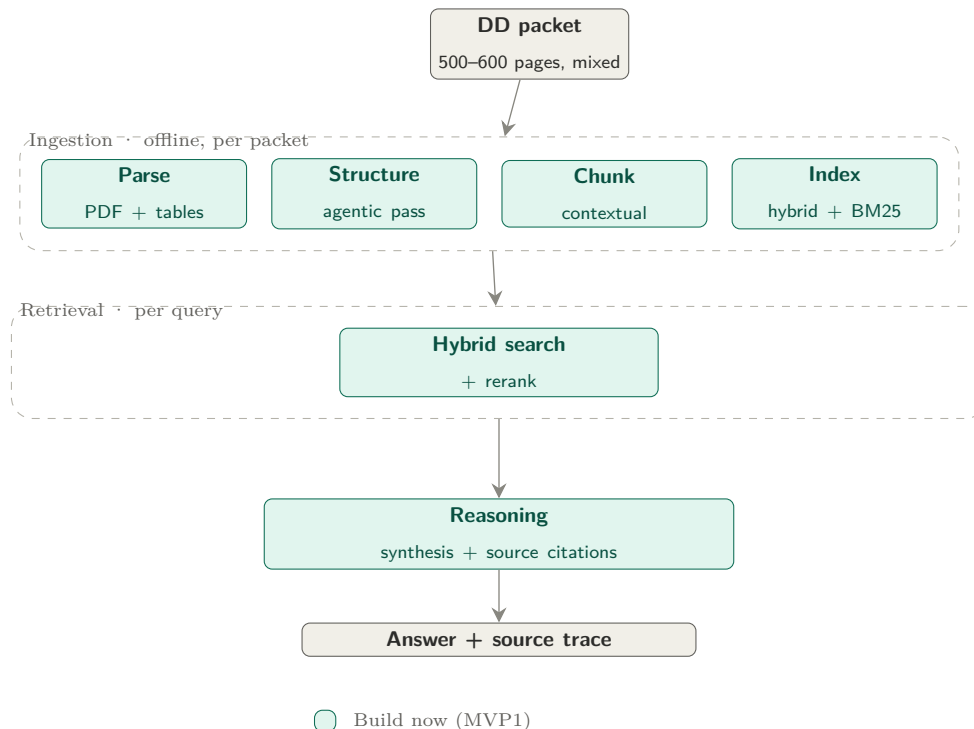
We are building an LLM-powered long-document extraction agent for private credit due diligence. It ingests 500–600 page DD packets - credit agreements, CIMs, financial statements, legal opinions - and extracts the information a credit analyst needs, with every answer traceable back to its source page. Because these are private companies, the input has no guaranteed structure, so the agent recovers structure first and extracts against it second.

The design is grounded in seven papers:

- (1) Lewis et al. (2020), *Retrieval-Augmented Generation* - the core retrieve-then-generate pattern.
- (2) Liu et al. (2023), *Lost in the Middle* - models use long context unevenly, so stuffing a whole document fails.
- (3) Yao et al. (2022), *ReAct* - interleave reasoning and retrieval in a loop.
- (4) Edge et al. (2024), *GraphRAG* - retrieve over an entity/relationship graph.
- (5) Lumer et al. (2025), *Rethinking Retrieval* - traversing document structure can match or beat vector RAG on financial documents.
- (6) Trivedi et al. (2022), *IRCoT* - interleave retrieval with chain-of-thought for multi-hop questions.
- (7) Dong et al. (2024), *SpreadsheetLLM* - how structured/tabular content is represented determines whether the model can use it.

Section 1 - MVP1

MVP1 is the full pipeline end to end with simple retrieval. The goal is a working baseline we can measure against the “model given good context” benchmark (CorpFin v2) before adding any retrieval sophistication.

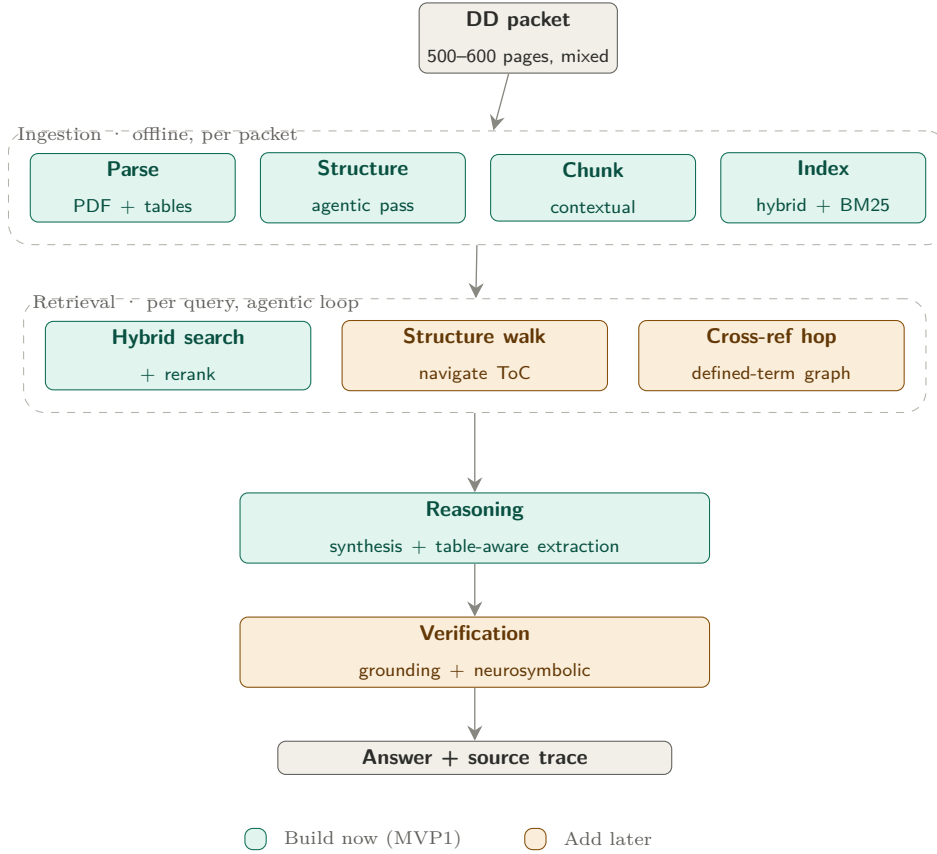


- **Parse** - layout-aware parsing of each PDF into clean text, tables, and per-element page coordinates. The coordinates are what give cell-level source traceability. (7) tells us to treat tables as structured objects, not flattened text, or numeric extraction later will fail.
- **Structure (agentic pass)** - an agent recovers the document's outline (sections, tables, entities) and the defined-terms glossary, since private-company packets carry no guaranteed structure. It produces a *map over* the original text and never rewrites it. (5) shows recovered structure is a strong retrieval signal; (4) motivates capturing the term/entity relationships.
- **Chunk** - chunk along the recovered structure boundaries and prepend each chunk with its location context (e.g. the section and definition it sits in). (2) is the reason we chunk-and-retrieve at all rather than stuffing 600 pages into the context window.
- **Index** - build a hybrid index combining dense embeddings with BM25 keyword search, plus metadata. (1) is the retrieval foundation; (5) shows hybrid search is effective on financial filings.
- **Retrieve (hybrid search + rerank)** - at query time, hybrid-search the index and rerank the top results. Single-pass in MVP1; (1) grounds the retrieve-then-generate flow.
- **Reason (synthesis + citations)** - the LLM answers from the retrieved chunks and always cites the source page. (1) grounds generation in retrieved evidence to reduce hallucination.
- **Output** - the answer plus its source trace, ready for analyst review.

MVP1 deliberately keeps retrieval simple. The point is to stand up the whole pipeline and get a measurable number before investing in the harder retrieval steps below.

Section 2 - MVP1 + later iterations

Iterations add capability exactly where the benchmark question types expose the MVP's weaknesses: multi-section cross-references, industry jargon, and numeric reasoning. The teal steps are unchanged; the amber steps are the additions.



- **Structure walk (retrieval)** - let the agent navigate the recovered outline / table of contents directly, not only by embedding similarity. (5) is the direct basis: structure traversal is competitive with, and on some query types beats, vector search on financial documents.
- **Cross-ref hop (retrieval)** - traverse the defined-term graph to resolve references that live far from the clause (e.g. a definition 200 pages away). (4) supplies the graph; (6) supplies the multi-hop reasoning chain that follows it.
- **Agentic loop** - turn retrieval into a reason → retrieve → re-retrieve loop that continues until it has enough to answer. (3) is the loop structure; (6) shows why a single-shot retrieval fails on multi-step questions.
- **Table-aware extraction (reasoning)** - handle financial tables as structured objects so numeric-reasoning questions resolve correctly. (7) is the basis.
- **Verification (grounding + neurosymbolic)** - check that each answer is grounded in its cited source span and is numerically consistent, catching the silent, confident errors generic agents make. This is our own differentiator, not drawn from the seven papers; it is motivated by benchmark findings that current agents fail without flagging it.

Deferred beyond this stage: domain-specific embedding fine-tuning, XBRL parsing, and multimodal (chart) retrieval. None belong in an early version.

References

1. Lewis, P. et al. (2020) “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” NeurIPS 2020. arXiv:2005.11401.
2. Liu, N. F. et al. (2023) “Lost in the Middle: How Language Models Use Long Contexts.” TACL 2024. arXiv:2307.03172.
3. Yao, S. et al. (2022) “ReAct: Synergizing Reasoning and Acting in Language Models.” ICLR 2023. arXiv:2210.03629.
4. Edge, D. et al. (2024) “From Local to Global: A Graph RAG Approach to Query-Focused Summarization.” arXiv:2404.16130.
5. Lumer, E. et al. (2025) “Rethinking Retrieval: From Traditional RAG to Agentic and Non-Vector Reasoning Systems in the Financial Domain for LLMs.” arXiv:2511.18177.

6. Trivedi, H. et al. (2022) “Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions (IRCoT).” ACL 2023. arXiv:2212.10509.
7. Dong, H. et al. (2024) “SpreadsheetLLM: Encoding Spreadsheets for Large Language Models.” arXiv:2407.09025.